Chapter 9

The Smoking Gun: Experimental Signatures

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othing would please string theorists more than to proudly present the world with a list of detailed, experimentally testable predictions. Certainly, there is no way to establish that any theory describes our world without subjecting its predictions to experimental verification. And no matter how compelling a picture string theory paints, if it does not accurately describe our universe, it will be no more relevant than an elaborate game of Dungeons and Dragons.

Edward Witten is fond of declaring that string theory has already made a dramatic and experimentally confirmed prediction: “String theory has the remarkable property of predicting gravity.”

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What Witten means by this is that both Newton and Einstein developed theories of gravity because their observations of the world clearly showed them that gravity exists, and that, therefore, it required an accurate and consistent explanation. On the contrary, a physicist studying string theory—even if he or she was completely unaware of general relativity—would be inexorably led to it by the string framework. Through its massless spin-2 graviton pattern of vibration, string theory has gravity thoroughly sewn into its theoretical fabric. As Witten has said, “the fact that gravity is a consequence of string theory is one of the greatest theoretical insights ever.”2 In acknowledging that this “prediction” is more precisely labeled a “postdiction” because physicists had discovered theoretical descriptions of gravity before they knew of string theory, Witten points out that this is a mere accident of history on earth. In other advanced civilizations in the universe, Witten fancifully argues, it is quite possible that string theory was discovered first, and a theory of gravity found as a stunning consequence.

Since we are bound to the history of science on our planet, there are many who find this postdiction of gravity unconvincing experimental confirmation of string theory. Most physicists would be far happier with one of two things: a bona fide prediction from string theory that experimentalists could confirm, or a postdiction of some